# MITS 5400G Secure Software Systems

# Winter 2026 - Mid-Term Exam Sample Questions

**Format: 50 Multiple Choice Questions**

**Scope: Week 1, 2 & 3**

**Closed book & No AI/device = ONLY a pen/pencil**

**Date: 13 February 2026 (Friday)**

**Time: 5:15 PM-6:45 PM (90 minutes)**

**1. What is the primary purpose of the layered structure in the THE multiprogramming system?**

A. To maximize execution speed by bypassing abstractions
B. To allow arbitrary interaction between system components
C. To reduce system complexity by enforcing strict abstraction boundaries
D. To eliminate the need for synchronization mechanisms

**Answer: C**

**2. In the THE system architecture, which rule governs interactions between layers?**

A. Any layer may invoke services from any other layer
B. Higher layers may use only the services of lower layers
C. Lower layers depend on higher layers for control
D. All layers execute independently without interaction

**Answer: B**

**3. What role do semaphores play in the THE multiprogramming system?**

A. They provide interprocess data exchange
B. They replace memory management mechanisms
C. They control access to critical sections and shared resources
D. They serve as user-level communication channels

**Answer: C**

**4. Which component enforces access control by mediating all access requests?**

A. Reference monitor
B. Access list
C. Kernel module
D. Session manager

**Answer:** A

**5. The main drawback of Discretionary Access Control (DAC) is that:**

A. It is too rigid
B. It allows uncontrolled information flow
C. It restricts user autonomy
D. It requires encryption

**Answer:** B

**6. Mandatory Access Control (MAC) is based on:**

A. User identity
B. Object ownership
C. Security classification levels
D. Encryption protocols

**Answer:** C

**7. Which statement best describes the primary limitation of the Waterfall software development model?**

A. It lacks formal documentation
B. It does not support modular design
C. It is inflexible to changing requirements after the early phases
D. It cannot be used for large projects

**Answer: C**

**8. Which practice is central to Extreme Programming (XP)?**

A. Formal change control boards
B. Writing code before tests
C. Pair programming and test-driven development
D. Eliminating user stories

**Answer: C**

**9. The CIA triad represents which three security objectives?**

A. Confidentiality, Integrity, Authentication
B. Confidentiality, Integrity, Availability
C. Control, Isolation, Authorization
D. Compliance, Integrity, Accountability

**Answer: B**

**10. According to the slides, which data state is the most difficult to protect?**

A. Data at rest
B. Data in transit
C. Data in use
D. Archived data

**Answer: C**